higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CARPENTRY AND ROOF WORK
NQF LEVEL 2

(12020122)

12 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages.

NCV

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NCV(V)

QUESTION 1: BASIC TOOLS AND TERMINOLOGY

1.1 Name a power tool with interchangeable blades that can be used to cut wood, metal and other materials. (1)

1.2 The block below contains a selection of carpentry tools and equipment. Choose any FIVE that can be used in roof construction.

sand paper; electric drill; chalk line; tape measure; plumb bob; spirit level; roofing square; straight edge; mitre box; claw hammer

(5)

1.3 Describe a plumb bob and explain its usage and purpose when erecting roof trusses. (11)

1.4 Give a brief description of the tool known as a sliding bevel/bevel square and say what it is used for. (3)

[20]

QUESTION 2: ROOF CONSTRUCTION

2.1 What type of tool was used by early man for cutting and shaping wood for building a shelter? (1)

2.2 Give the TWO main functions of a roof. (2)

2.3 Differentiate between a *dead load* and a *live load* on a roof, and give TWO examples of each type. (6)

2.4 Name FIVE different types of roof structures and give a brief description of each one. (5 × 2) (10)

2.5 Differentiate between a *rationally designed roof* and a *deemed-to-satisfy roof.* (2 × 2) (4)

2.6 Name FOUR principles applied in roof construction. (4)

2.7 Explain the function of the wall plate and say where it is usually fixed. (2)

2.8 SANS 102443 gives specific guidelines with regard to the dimensions of wall plates.

Give the correct dimensions of the wall plates for the following roof spans:

2.8.1 8 meters

2.8.2 12 meters

2.8.3 19 meters

(3 × 1) (3)

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2.9 Give the maximum permissible distance that roof anchors may be fixed away from a roof truss. (1)

2.10 Differentiate between *battens* and *purlins* and give ONE function that each serves in the roof structure. (4)

2.11 Name TWO devices that cranes make use of to ensure the safe raising of heavy trusses. (2)

2.12 Explain the purpose of the washer when using bolts during roof truss construction. (1)

[40]

QUESTION 3: BUILD A HOWE TRUSS

3.1 Choose a term from COLUMN A that matches a description in COLUMN B. Write only the letter (A–F) next to the question number (3.1.1–3.1.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.1.1	Run	A	Pythagoras' theorem used to measure a 90˚ angle
3.1.2	Clear span	B	positioned vertically in the truss
3.1.3	3:4:5 method	C	height of the roof measured from the underside of two rafters at the apex to the bottom of tie-beam
3.1.4	King post	D	half the tie beam
3.1.5	Rise	E	the inside distance between the supporting walls
		F	where two rafters meet at the very top of the truss

(5 × 1) (5)

3.2 Name the THREE stages of constructing a truss. (3)

3.3 Give a brief explanation of the following terms commonly used when setting out a roof truss:

3.3.1 Node

3.3.2 Pitch

3.3.3 Heel

3.3.4 Apex

(2 × 4) (8)

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3.4 Determine the rise of the roof truss if the pitch is 30^0 and tie beam span is 4 500 mm. (3)

3.5 Which member of the roof truss will receive *plumb cuts*? (1)
[20]

QUESTION 4: HOW TO BUILD A GABLE ROOF

4.1 Give TWO safety precautions to be observed on a building site. (2)

4.2 Name the TWO places on a roof where the carpenter can start erecting the roof trusses. (2)

4.3 Name THREE ways of preventing damage and injury when handling roof trusses. (3)

4.4 Explain what is meant by the term *roof cladding.* (2)

4.5 Give the reason why three workers are needed to level and align roof trusses. (2)

4.6 Name THREE things that can be placed between the truss and the wall to level the truss. (3)

4.7 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (4.7.1–4.7.6) in the ANSWER BOOK.

4.7.1 A roof anchor is a tie wire or iron strap that is used to tie down the truss.

4.7.2 The minimum size of permanent bracing timber is 38 mm x 50 mm for spans up to 6, 6 metres.

4.7.3 In shops and offices, there must be at least one health and safety representative for every 50 employees.

4.7.4 The type of roof covering strongly influences the batten and purlin sizes and also affects truss spacing.

4.7.5 Wall plates are always supported on the outermost skin of brickwork.

4.7.6 The *tilting batten* supports the roof gutters.

(6 × 1) (6)
[20]

TOTAL: 100

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